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Reconstructing Architectural Heritage through 3D and Immersive Technologies: Integrating Intangible Practices for Authentic Representation—A Transferable Framework from the Gedi Ruins Case Study

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Note: This paper builds on research first undertaken as part of an MSc dissertation at the Edinburgh School of Architecture and Landscape Architecture (ESALA), University of Edinburgh, and further developed through ongoing PhD research at the University of Liverpool.

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Abstract

This research investigates how augmented reality (AR), virtual reality (VR), and web-based 3D visualisation can support the authentic interpretation and transmission of intangible architectural heritage. While digital heritage technologies have advanced the documentation and reconstruction of physical remains, they often fail to capture the social practices, embodied knowledge, and cultural meanings that underpin architectural production and use. Consequently, digital reconstructions risk presenting heritage sites as static artefacts disconnected from the communities and traditions that give them significance, raising ongoing questions surrounding authenticity, cultural representation, and ethical interpretation within immersive heritage environments (UNESCO, 2003).

Using the Gedi Ruins in Kenya as a case study, the research explores how web-based 3D visualisation, AR, and VR environments have been used to support both spatial reconstruction and the transmission of intangible architectural knowledge within Swahili cultural contexts. The study employs a qualitative methodology that combines field observations, semi-structured interviews, documentary analysis, and user engagement feedback. Data was collected from ten experts and community contributors, including representatives from the National Museums of Kenya, a UNESCO-affiliated heritage professional responsible for the Gedi site, members of the Zamani Project involved in the site's 3D documentation, and community stakeholders, such as a former Mayor of Malindi. The study also analysed transcribed video interviews and presentations by heritage specialists as primary data sources. These methods were complemented by 3D spatial documentation and immersive digital reconstruction workflows designed to capture vernacular building techniques, spatial organisation, symbolic meaning, and oral transmission practices embedded within Swahili architectural traditions.

Rather than developing a new 3D reconstruction workflow, the study critically evaluates existing digital documentation of the Gedi Ruins produced by the Zamani Project using photogrammetry and 3D modelling techniques. The analysis compares the architectural information represented within these digital reconstructions against evidence of Swahili building practices, construction processes, spatial organisation, and culturally embedded knowledge gathered through interviews and documentary sources. This comparison identifies gaps between the documentation of physical architectural features and the representation of the intangible knowledge systems that informed their creation and use. Findings suggest that while existing 3D reconstructions provide valuable tools for visualisation and documentation, they are less effective as learning environments for communicating architectural processes, embodied knowledge, and cultural meaning. Interpretive depth increases when digital environments incorporate storytelling, cultural context, and community-informed interpretation alongside visual reconstruction.

In response, the study proposes a transferable framework for integrating intangible cultural heritage into immersive visualisation systems. The framework can be adapted for use at other heritage sites by combining community participation, storytelling, cultural interpretation, and immersive technologies, providing a practical methodology that can be adapted by museums and heritage organisations seeking to integrate local knowledge into immersive heritage experiences, propagating authentic representation. It also provides guidance for evaluating how effectively digital heritage experiences communicate cultural knowledge and engage users. While grounded in the Gedi Ruins case study, the framework is designed for broader application across diverse heritage sites, museums, and digital heritage platforms, offering scalable guidance for institutions seeking to represent architectural heritage as a lived and culturally situated process rather than a static object.

The framework further supports FAIR-aligned heritage practices by supporting long-term preservation, accessibility, and reuse of heritage knowledge through clear documentation and community involvement. By bridging technical reconstruction with cultural representation, this research contributes to ongoing debates surrounding ethics, authenticity, public engagement, and digital preservation within the field of heritage practice. It also provides practical guidance for developing immersive heritage tools that can communicate both the tangible and intangible dimensions of architecture, while supporting endangered knowledge systems and community-centred heritage preservation (London Charter, 2009; Bekele et al., 2018; Champion, 2016).

References

1. Bekele, M.K., Pierdicca, R., Frontoni, E., Malinverni, E.S. and Gain, J. (2018). A survey of augmented, virtual, and mixed reality for cultural heritage.
2. Journal of Cultural Heritage, 31, pp.1–24. Champion, E. (2016). Critical Gaming: Interactive History and Virtual Heritage.

3. Routledge. Denard, H. (2012). A New Introduction to the London Charter. In: Paradata and Transparency in Virtual Heritage.
4. London Charter (2009). The London Charter for the Computer-based Visualisation of Cultural Heritage.
5. UNESCO (2003). Convention for the Safeguarding of the Intangible Cultural Heritage.
6. Kenderdine, S. (2013). Immersive visualization for cultural heritage. In: Digital Heritage International Congress.